Stock Market Clustering (Unsupervised Learning)

Machine Learning project · Adewale Adeyinka Falade · Python, scikit-learn, scipy

Problem

Given financial attributes for ~340 large-cap (S&P 500) companies, discover natural groupings of stocks that behave similarly — useful for diversification, risk management and portfolio construction.

Approach

- ~340 stocks: price, volatility, ROE, cash ratio, net income, EPS, P/E, P/B, sector.
- EDA on P/E and volatility by sector; outlier checks; feature scaling.
- K-means clustering — k chosen via elbow method and silhouette analysis.
- Hierarchical (Agglomerative) clustering — cophenetic correlation across linkage methods; dendrograms.
- Compared both techniques on cluster distinctness, membership and suggested cluster count.

Results

- Final K-means model segmented stocks into **3 clusters** (silhouette ≈ 0.40 at k = 3).
- **Financial-sector stocks** (major banks) formed their own distinct cluster — different ROE, leverage and valuation profile.
- Hierarchical clustering gave broadly consistent groupings, supporting the K-means structure.

Why It's Useful

The clusters give an investor a map of which stocks behave alike. Diversifying *across* clusters — rather than holding many names from the same cluster — spreads risk more effectively than sector labels alone.

Self-authored notebook. Course brief not redistributed; dataset is public S&P 500 financial data.